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Review

Israel Journal of Chemistry

DOI: 10.1002/ijch.201700034

Transition Metal-Catalyzed Benzannulation towards Naturally Occurring Carbazole Alkaloids

Xue Zhang[a] and Shengming Ma[a, b]

Abstract: In this account, we will summarize our recent progress on development of Pt- or Au-catalyzed benzannulation of 1-indolyl-2,3-allenols for the efficient construction of carbazole skeleton via a metal carbene intermediate and its application towards the synthesis of naturally occurring carbazole alkaloids. The strategy enjoys the outstanding regioselectivity, atom economy, mild reaction conditions, substrate scope, functional group tolerance.

Keywords: Gold-catalysis · Benzannulation · Carbazole alkaloids · Metal carbenes · Total synthesis

1. Introduction

Carbazole alkaloids have received increasing attention as an important class of heteroaromatic compounds due to their unique biological and pharmacological activities, as well as their exceptional thermal, electrical, and optical properties.[1] During the last four decades, an increasing number of naturally occurring carbazole alkaloids have been isolated from different natural sources, among which the taxonomically similar plants of the genus *Murraya*, *Glycosmis*, *Clausena* and *Micromelum* from the family *Rutaceae* represent the richest sources.[2] Their pharmacological effects are very broad, including *anti*-HIV, *anti*-TB, *anti*-malarial, *anti*-tumor, and neuronal cell protective activity.[1]

The biologically active carbazole alkaloids call for the diversity of compounds with such a core structure. Thus, many synthetic approaches have been developed.[3] The methodologies that have been often utilized for the construction of carbazoles skeleton are summarized as follows: the Fischer-Borsche synthesis is one of the classical procedures for carbazole alkaloids (Scheme 1a);[4] condensation of cyclohexanone with phenylhydrazines affords the arylhydrazones, which could cyclize to afford the indole unit through Fischer-Borsche cyclization. Then, dehydrogenation of the tetrahydrocarbazoles produces carbazoles finally by using palladium on activated carbon or chloranil (Scheme 1a); another widely used method for carbazole synthesis is the deoxygenative cyclization of *o*-nitrobiphenyls in triethyl phosphite under reflux, which is also known as Cadogan synthesis (Scheme 1b).[5] 1-Phenylbenzotriazole, prepared by diazotization of N-(2-aminophenyl)aniline, would transform to carbazole under thermal reaction conditions, which is called Graebe-Ullmann synthesis (Scheme 1c).[6] Knölker et al. reported iron-mediated carbazole synthesis protocol (Scheme 1d):[7] The electrophilic aromatic substitution of arylamines by the tricarbonyl iron-coordinated cyclohexadienylium ions affords an arylamine-substituted tricarbonyl(η^4-cyclohexadiene)iron complex, which would lead to a carbazole by oxidative cyclization; cyclodehydrogenation of diarylamines is another practical method for the synthesis of carbazoles. Diarylamines can be obtained directly by palladium(0)-catalyzed Buchwald-Hartwig coupling of aryl halides with arylamines (Scheme 1e).[8] Similarly, carbazoles may also be synthesized via 2-fold palladium(0)-catalyzed Buchwald-Hartwig coupling of biphenyl-2,2′-diyl bistriflates (or 2,2′-dihalobiphenyls) with primary amines (Scheme 1f);[9] Witulski et al. reported a creative approach for carbazole synthesis by employing a rhodium-catalyzed [2+2+2] cyclotrimerization of a diyne and an alkyne as the key step (Scheme 1g).[10]

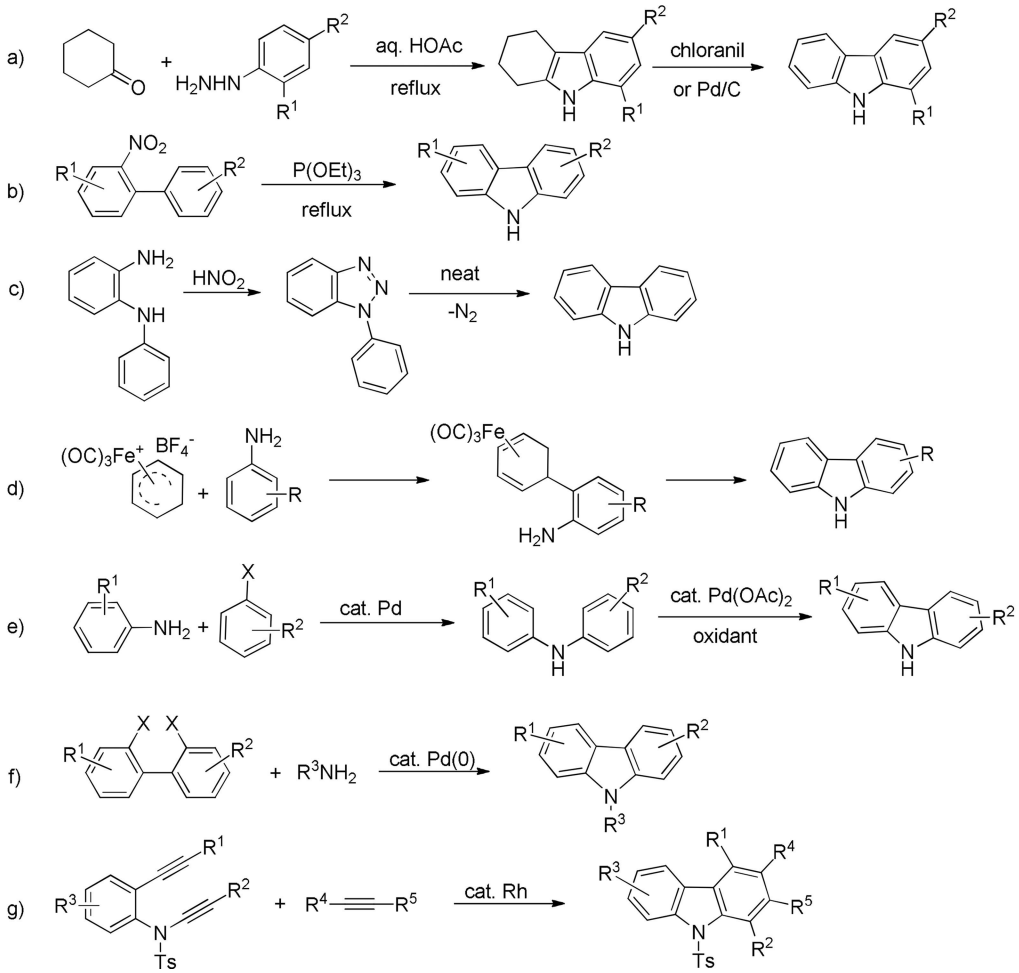
The above mentioned synthetic strategies for carbazoles are mostly based on construction of the central pyrrole ring by cyclization of either diarylamines or biphenyls with an ortho-nitrogen substituent. Forming a benzene ring to indole via electrocyclic reaction is another effective strategy for carbazole construction. The wide availability of indole derivatives makes this strategy quite practical. A large number of carbazole derivatives have been prepared efficiently by electrocyclic reaction of 2,3-divinylindoles at high temperature in the presence of dehydrogenating agents such as Pd/C or DDQ (Scheme 2a).[11] An improvement of this electrocyclic procedure is cyclization of 2,3-difunctionalized indole via an allene intermediate (Scheme 2b).[12] A further indole benzannulation of 3-(buta-1,3-dienyl)indoles has been applied to the synthesis of 3-oxygenated carbazoles (Scheme 2c).[13] In addition to the methods described above, there are still some alternative protocols.[3]

Although numerous effective synthetic approaches for the construction of carbazole framework have been developed, there remain some limitations,[3] such as low yields, harsh

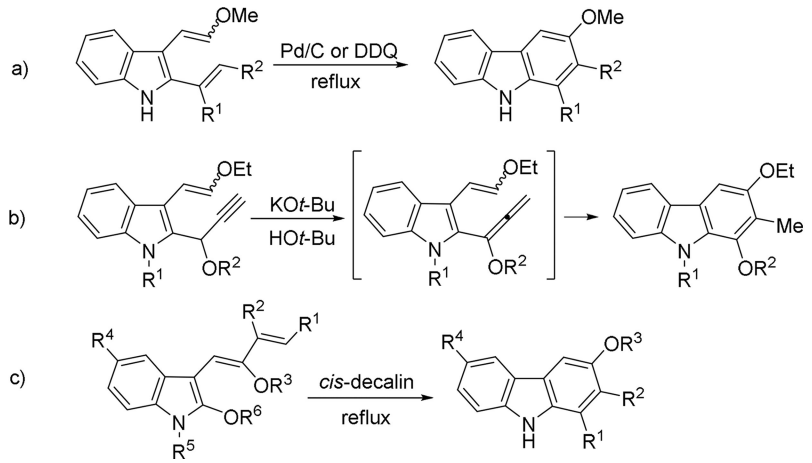
[a] X. Zhang, S. Ma
State Key Laboratory of Organometallic Chemistry, Shanghai Institute of Organic Chemistry, Chinese Academy of Sciences, 345 Lingling Lu, Shanghai 200032, P.R. China
[b] S. Ma
Research Center for Molecular Recognition and Synthesis, Department of Chemistry, Fudan University, 220 Handan Lu, Shanghai 200433, P.R. China

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Scheme 1. The methodologies for carbazole skeleton construction.



Scheme 2. Carbazole skeleton construction by forming a benzene ring to indole.

reaction conditions, poor atom or step economy, and high loading of expensive/dangerous/toxic reagents. Thus, the development of mild, efficient, practical, and general methods for carbazole framework construction from easily available starting materials is still highly desirable. In 2009, we developed an efficient approach for the construction of

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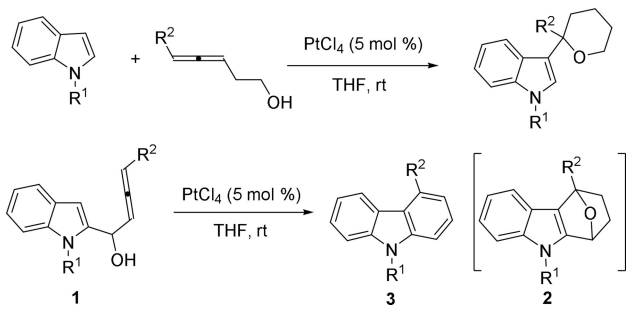
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carbazole skeleton through Pt-catalyzed cyclization of 1-(indol-2-yl)-2,3-allenols.[14] The reaction occurred smoothly at room temperature to afford a series of poly-substituted carbazoles. Two years later, our further investigation led to the observation of AuCl as a much more effective catalyst for this transformation.[15] Due to the easy availability of the starting materials, this simple and mild Au-catalyzed reaction of 1-(indol-2-yl)-2,3-allenols provides a general route to differently substituted carbazoles. In this account, we will summarize the progress we made on this topic beginning with the methodology development, followed by the application of the current strategy in the (gram scale) total synthesis of a serial of naturally occurring carbazole alkaloids.

2. Reaction Development

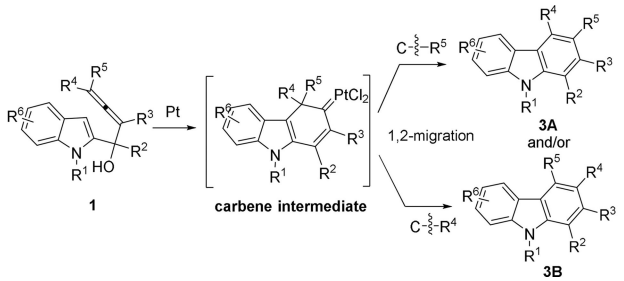
2.1 [$PtCl_2$]-Catalyzed Cyclization of 1-(Indol-2-yl)-2,3-allenols

Intrigued by our previous observation of $PtCl_4$-catalyzed intermolecular cyclization reaction of indoles with β-allenols, which afforded indole derivatives with a saturated 6-membered cyclic ether at the C3-position (Scheme 3),[16] we envisioned that polycyclic products **2** would be obtained if 1-(indol-2-yl)-2,3-allenols **1** undergo a similar reaction. However, carbazoles **3** were formed unexpectedly at room temperature (Scheme 3). The starting materials are easily available from the related 1*H*-indole-2-carbaldehydes based on known methods.[17] After careful screening of the reaction parameters, the standard reaction conditions are defined as follows: the reaction of 1-(indol-2-yl)-2,3-allenols **1** was conducted with 5 mol% $PtCl_2$ in toluene at room temperature. This transformation is quite general with a very good functional group tolerance. Some typical results are listed in Table 1.[14]



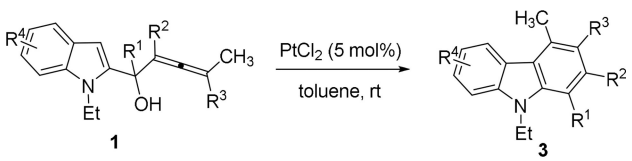
Scheme 3. $PtCl_4$-catalyzed cyclization reaction of indolyl-2,3-allenols.

A possible mechanism involving a 1,2-H migration of the Pt-carbene intermediate was proposed for this transformation. In addition, 4,4-carbon disubstitution was introduced to the 1-(indol-2-yl)-2,3-allenols to further check the intermediacy of the Pt-carbene intermediate and the selectivity of the subsequent carbon-carbon single bond cleavage via 1,2-carbon migration for formation of **3A** or **3B** (Scheme 4).[18]



Scheme 4. Proposed mechanism for the 1,2-carbon migration of the Pt-carbene intermediate.

After some trials, we achieved a highly selective carbon-carbon single bond cleavage reaction of 4-aryl-4-alkyl-substitued allenols. Different carbazoles were provided in moderate to good yields with the aromatic group migration exclusively. Further introduction of different substituents to the original indole substrate shows that this protocol provides an efficient approach to assemble the substituents to any location of carbazole skeleton (Scheme 5).[18]



R^1 = Me, R^2 = H, R^3 = Ph, R^4 = H	83%
R^1 = H, R^2 = Me, R^3 = Me, R^4 = H	55%
R^1 = H, R^2= H, R^3 = *p*-MeC_6H_4, R^4 = 5-Me	83%
R^1 = H, R^2 = H, R^3 = Me, R^4 = 5-OMe	70%
R^1 = H, R^2 = H, R^3 = Me, R^4 = 7-Me	72%

Scheme 5. $PtCl_2$-catalyzed cyclization reaction of 4,4-disubstitued 1-(indol-2-yl)-2,3-allenols **1**.

2.2 [AuCl]-Catalyzed Cyclization of 1-(Indol-2-yl)-2,3-allenols

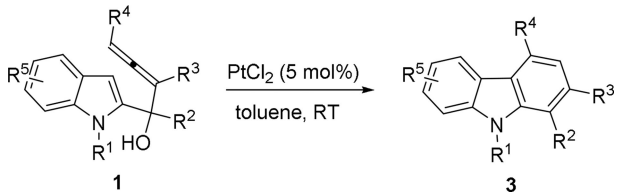
For the purpose of synthesizing the naturally occurring carbazole, a methoxy group has been introduced to the 2-position of the starting 1-(indol-2-yl)-2,3-allenols. We observed that AuCl is a much more effective catalyst than $PtCl_2$.[15] In the total synthesis of siamenol, we utilized this transformation as the key step for constructing the carbazole skeleton from the required methoxy-substituted allenol **1a**. Under the standard reaction conditions,[14] the $PtCl_2$-catalyzed cyclization of **1a** provided carbzaole **3a** in a relatively low yield (61%), probably due to the existence of methoxy group. However, when AuCl was used as the catalyst, **3a** was produced in a much higher yield (81%), exclusively (Equation 1).[15]

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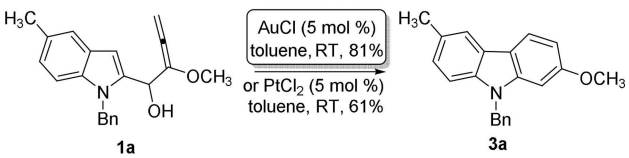
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Table 1. $PtCl_2$-catalyzed cyclization reaction of 1-(indol-2-yl)-2,3-allenols.[14]



1 3

Entry	R^1	R^2	R^3	R^4	R^5	Time/h	Yield of **3** (%)
1	Et	H	H	*n*-C_5H_{11}	H	2	83
2	Et	H	H	Me	H	2	86
3	Et	H	H	*n*-C_6H_{13}	5-Me	2	75
4	Et	H	H	*n*-C_6H_{13}	5-OMe	3	75
5	Et	H	H	Bn	H	5	50
6	Me	H	Ph	H	H	4	81
7	Me	H	*n*-C_4H_9	H	H	4	71
8	Et	H	Allyl	H	H	3	74
9	Ph	H	Ph	H	H	19	78
10	Et	H	COOMe	H	H	36	67
11	Et	H	$CONMe_2$	H	H	24	83
12	Et	H	CH_2OH	H	4-Me	20	81
13	Et	H	CH_2OH	H	7-Me	21	68



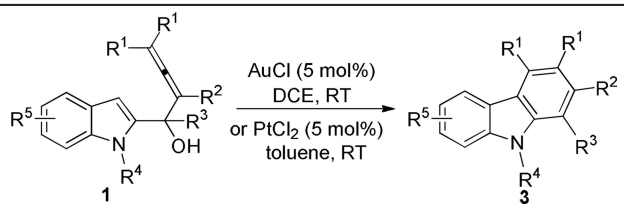
1a 3a

Moreover, with AuCl as the catalyst, cycloisomerization of 4,4-dimethyl substituted allenols also proceeded smoothly to afford better yields of carbazoles in a shorter time than the previous $PtCl_2$-catalyzed process (entries 1–3, Table 2).[19] In addition to methyl group, R^1 may also be ethyl and propyl group (entries 5 and 6, Table 2).

Furthermore, we investigated the selectivity of 1,2-migration in this transformation to check the order of migration: compared with methyl group, isopropyl, cyclopropyl, cyclobutyl, and cyclohexyl migrate exclusively; cyclopropyl group shifts selectively over ethyl group; the 1,2-migration of a non-methyl linear alkyl is faster than methyl group; the phenyl group migrates exclusively over methyl or ethyl group.[19] The AuCl-catalyzed reaction represents a significant advance in terms of the reactivity, the scope, and the selectivity, as compared with the [$PtCl_2$]-catalyzed process.[19]

A plausible mechanism was proposed according to the deuterium-labeling experiment, which was further confirmed by DFT calculations (Scheme 6).[19] The initial formation of a complex **IM1** involves the coordination of $PtCl_2$ to the distal allenic double bond of **1**, which would be followed by nucleophilic attack of the indolyl C3. This *6-endo-trig*

Table 2. AuCl- vs. $PtCl_2$-catalyzed cyclization reaction of 4,4-dialkyl-1-(indol-2-yl)-2,3-allenols.



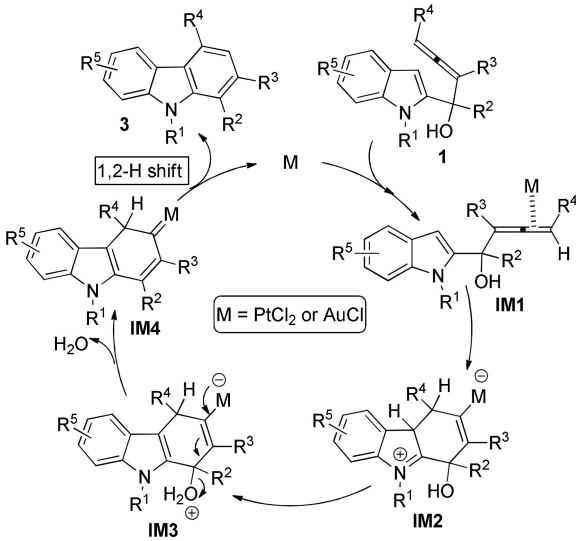
1 3

Entry	$R^1/R^2/R^3/R^4/R^5$	Yield/Time[a] for [AuCl] (%/h)	Yield/Time[18] for [$PtCl_2$] (%/h)
1	Me/H/H/Et/H	88/1	73/10
2	Me/H/H/Et/7-Me	85/1	72/16
3	Me/Me/H/Et/H	69/1	55/12
4	Me/H/Me/Et/H	87/1	–
5	Et/H/H/Bn/H	57/4[b]	–
6	Pr/H/H/Bn/H	34/24[b,c]	–

[a] *Reaction conditions:* The reaction was conducted with 0.2 mmol **1** and AuCl (5 mol%) in 10 mL of DCE under N_2; [b] DCE (20 mL), [c] AuCl (15 mol%) was used.

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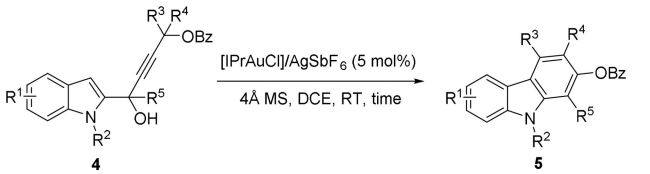
Scheme 6. The mechanism confirmed by DFT calculations.

cyclization step provides the zwitterionic cyclic species **IM2**. Subsequent proton abstraction and protonation of the hydroxyl group are assisted by the *in-situ* generated water to form intermediate **IM3**. Then the elimination of H_2O from **IM3** affords the platinum carbene intermediate **IM4**. 1,2-H-shift reaction of the platinum carbene intermediate **IM4** produces the final product **3** with regeneration of the catalytically active species $PtCl_2$. According to our computational studies, the mechanism of the AuCl-catalyzed process is similar to that of [$PtCl_2$], but less energy demanding in the rate-limiting 1,2-H shift step. It may be due to the higher ability of Pt compared with Au for forming and stabilizing carbenic C=M bonds that denotes the relative stability of Pt-carbene **IM4**, hence, showing a lower reactivity for the subsequent 1,2-H-migration. The migratory order of different alkyl or phenyl groups was also validated by the DFT calculations.[19]

2.3 [IPrAuCl]/$AgSbF_6$-Catalyzed Cyclization of 4-benzoxyl-1-(indol-2-yl)-2-alkynols

In 2014, another efficient procedure for carbazole framework construction from indoles is developed by our group.[20] [IPrAuCl]/$AgSbF_6$-catalyzed cyclization of 4-benzoxyl-1-(indol-2-yl)-2-alkynols **4** occurred smoothly in 1,2-dichloroethane (DCE) in the presence of 4 Å MS, producing a series of differently poly-substituted 2-oxygenated carbazole derivatives in good yields under very mild conditions (Table 3). The reaction was proven to be quite broad in substrate scope (Table 3).[20] Highly selective migration is also observed in this transformation: compared with methyl group (R^3), propyl, 4-methylphenyl, and 4-chlorophenyl groups (R^4) all migrates exclusively (Table 3, entries 10–13). Particularly, the starting materials are readily available from the reaction of indolecarbaldehydes with the related terminal alkynes. With this

Table 3. Gold-catalyzed cyclization reaction of 4,4-disubstitued-4-benzoxyl- 1-(indol-2-yl)-2-alkyn-1-ols **4**.[a]



Entry	$R^1/R^2/R^3/R^4/R^5$	Time (h)	Yield of **5**(%)[b]
1	H/Bu/CH_3/CH_3/H	2	81
2	H/allyl/CH_3/CH_3/H	4	52
3	H/Ph/CH_3/CH_3/H	5	31
4	H/Bn/CH_3/CH_3/H	4	50
5	H/Et/CH_3/CH_3/Bu	2	90
6	H/Et/CH_3/CH_3/Ph	2	89
7	5-CH_3/Et/CH_3/CH_3/H	3	82
8[c]	5-CH_3/Et/CH_3/CH_3/H	3	86
9	H/Et/-$(CH_2)_3$-/H	3	72
10[d]	H/Et/CH_3/Pr/CH_3	4	88
11[e]	5-OMe/Et/CH_3/Pr/Bu	4	85
12[f]	H/Et/CH_3/4-MeC_6H_4/H	24	44
13[f]	H/Et/CH_3/4-ClC_6H_4/H	48	55

[a] The reaction was conducted with 1.0 mmol of **4**, 5 mol% [IPrAuCl]/$AgSbF_6$, and 4 Å MS in 10 mL of DCE at room temperature; [b] Isolated yield; [c] The reaction was conducted on 3.0 mmol scale; [d] The ratio was 96:4; [e] The ratio was 97:3; [f] The reaction was conducted at 80 °C.

protocol, poly-substituted carbazols could be obtained by assembling substituents to the phenyl ring, the N-atom of the indole unit, and the 1-position and two 4-positions of the alkynols.

A plausible mechanism is proposed based on the deuterium-labelling experiment (Scheme 7).[20] The benzoate unit in substrate **4** undergoes 1,3-migration, via the nucleophilic attack of the carbonyl oxygen on the Au^+-coordinated C–C triple bond, generating the allene intermediate **IM6**. *6-Endo-trig*-type nucleophilic attack of the indolyl C3 at the coordinated C=C bond in the allene moity would generate the vinyl–gold species **IM7**. Subsequent elimination of H_2O affords a gold-carbene intermediate **IM8**. The final 1,2-R^4 (R^4 = H, aryl, and alkyl) migration of **IM8** would produce the 2-oxygenated carbazoles **5** with regeneration of the gold catalyst (Scheme 7).

3. Total Synthesis

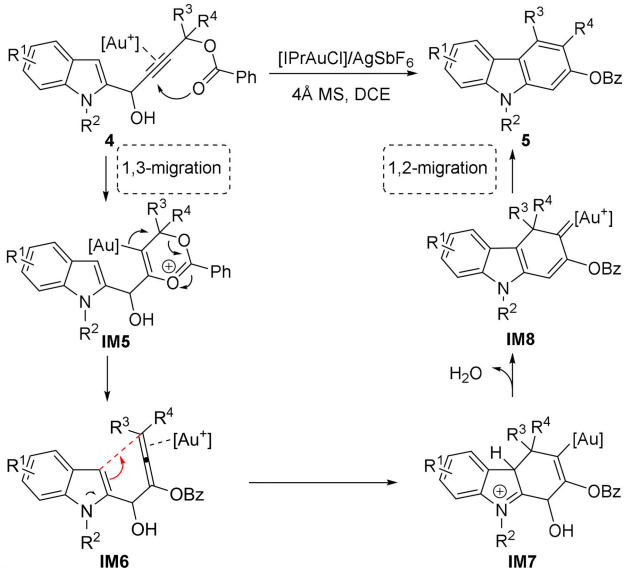
3.1 Total Synthesis of Naturally Occurring 2-Oxygenated Carbazole Alkaloids

As mentioned above, 2-oxygenated tricyclic carbazole alkaloids have attracted considerable attention owing to their biological activities (Scheme 8).[1] For example, many 2-oxygenated carbazole alkaloids have been isolated from different parts of *Murraya* and *Clausena*, which have been

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Scheme 7. Proposed mechanism.

claimed to be useful as herbal medicine traditionally.[2] Actually, most of these representative isolated compounds indeed show useful biological activity (Scheme 8).

Based on the Au-catalyzed approach to 2-oxygenated carbazoles, siamenol, 3-formyl-7-hydroxycarbazole, clauszoline-K, clauszoline-L, clausine-M and clausine-N were synthesized.[15] In 2013, we further applied this protocol to the general synthesis of isomukonidine, clausine L, mukonidine, glycosinine, mukonal, and clausine V.[21] To further demonstrating the great potential for its application as a general approach for the synthesis carbazole alkaloids, we realized the gram scales synthesis of 7-methoxy-O-methylmukonal, clausine-O, clausine-K, clausine-H, 7-methoxymukonal, methyl 2-hydroxy-7-methoxy-9H-carbazole-3-carboxylate,[22] 2-hydroxy-3-methylcarbazole, pyrano[3,2-*a*]carbazole alkaloids (girinimbine and murrayacine) and prenylcarbazole alkaloids (mukoenine-A and heptaphylline).[23] Here, we describe the total synthesis of these carbazole alkaloids in detail to show the beauty of our newly developed methodologies.

By retrosynthetic analysis, the Au-catalyzed cyclization of 1-indoly-2-methoxy-2,3-allenols is the key step (Scheme 9).[15] Readily available methoxypropadiene[24] and indole-2-carbaldehyde are chosen as the starting materials for the synthesis of these starting indoly-2,3-allenols.[17]

3.1.1 Total Synthesis of Siamenol

The required methoxy-substituted allenol **1a** may be synthesized conveniently from the addition of 1-methoxypropa-1,2-dienyl lithium[24] to 1-benzyl-5-methylindole-2-carbaldehyde **6a**. Subsequently, AuCl-catalyzed cyclization of **1a** under the standard reaction conditions afforded carbzaole **3a** exclusively in a good yield. Regioselective electrophilic bromination of **1a** at C-3 with NBS provided **7a**. The following debenzylation using potassium *tert*-butoxide/DMSO under oxygen atmosphere[25] produced **8a**. Then, **9a** would be formed by the demethylation of **8a** with BBr_3 in CH_2Cl_2.[26] When we used acetate as the protection form of the free OH group (**10a**), siamenol was eventually obtained via the coupling and instant deprotection in one-pot (Scheme 10).[15] Total synthesis of siamenol from methoxypropadiene is 7 steps with 21% overall yield.

3.1.2 Total Synthesis of Isomukonidine, Clausine L, Mukonidine, Glycosinine, and Mukonal

From 1-benzyl-1H-indole-2-carbaldehyde **6b,** a similar approach may also be used for the synthesis of carbazole **8b**. Starting from **8b**, isomukonidine was obtained by Br–Li exchange with *n*-BuLi and the subsequent carboxylation with CO_2; clausine L was provided by esterification of the carboxylic acid functionality in isomukonidine with MeI; mukonidine was synthesized by removal of the methyl group in the methoxy group in clausine L by treatment with BBr_3 in CH_2Cl_2;[26] glycosinine was generated by deprotonation of **8b** with KH, Li–Br exchange, and addition of dimethylformamide; mukonal was afforded by demethylation of glycosinine by treatment with BBr_3 in CH_2Cl_2 (Scheme 11). The net results for the total synthesises of these carbazole alkaloids from methoxypropadiene are: isomukonidine (5 steps with 29% overall yield), clausine L (6 steps with 26% overall yield), mukonidine (7 steps with 17% overall yield), glycosinine (5 steps with 34% overall yield), and mukonal (6 steps with 30% overall yield).[21]

3.1.3 Total Synthesis of Clausine V

A similar approach was also used for the synthesis of allenol **1c**, which underwent the AuCl-catalyzed cyclization smoothly under the standard reaction conditions to afford carbazole **3c,** exclusively. Clausine V was provided by debenzylation with potassium *tert*-butoxide/DMSO under oxygen atmosphere (Scheme 12). Total synthesis of clausine V from methoxypropadiene took 3 steps with 61% overall yield.[21]

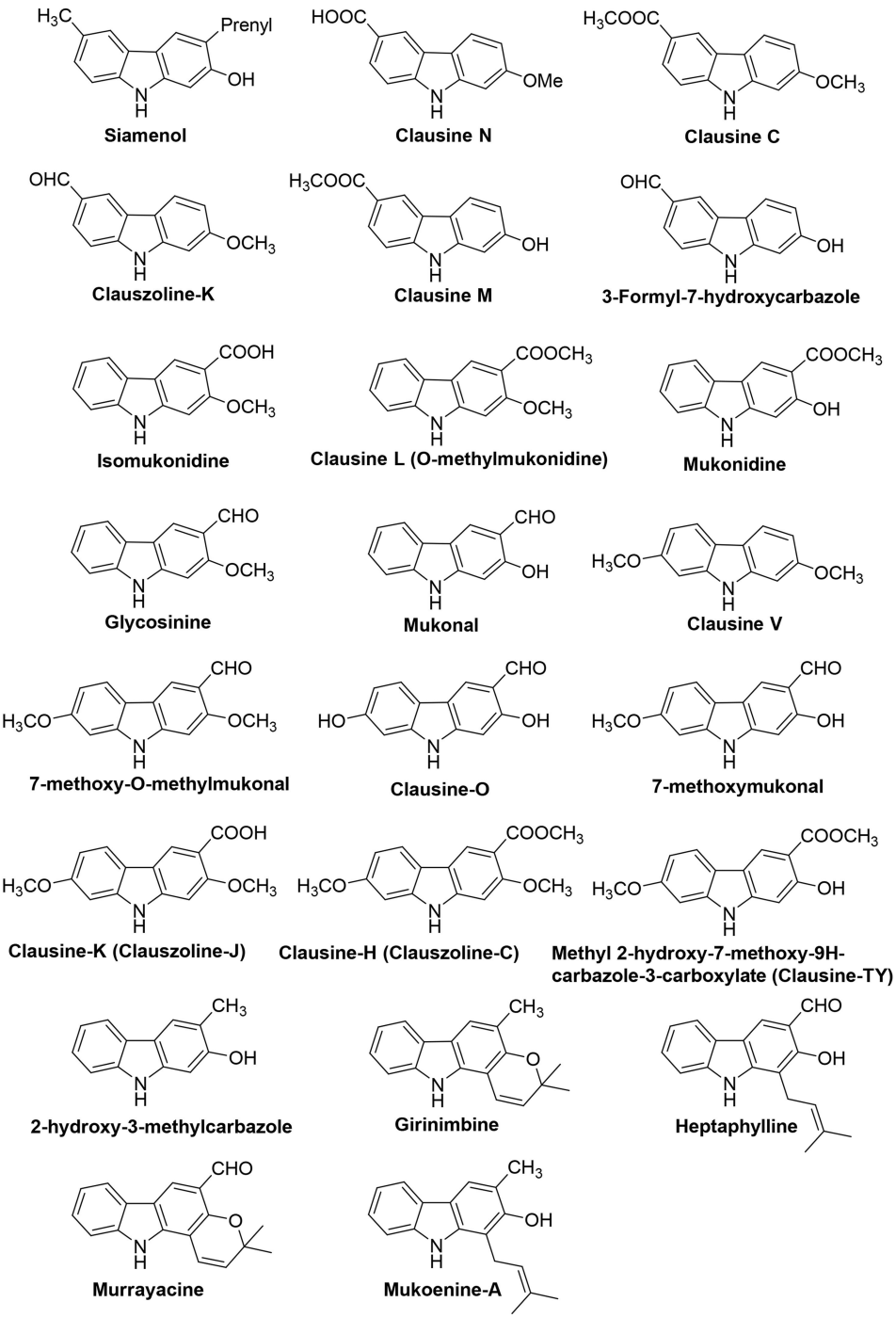
3.1.4 Total Synthesis of 3-Formyl-7-hydroxycarbazole, Clauszoline-K, Clauszoline-L, Clausine-M, and Clausine-N

This approach was used again for the synthesis of allenol **1d**, which underwent the AuCl-catalyzed cyclization smoothly under the standard reaction conditions to afford carbazole **3d** exclusively. Debenzylation with potassium *tert*-butoxide/DMSO under oxygen atmosphere[25] provided **11d**. Starting from **11d**, Clausine-N was produced by using the Li–Br exchange reaction and the subsequent carboxylation. Clauszoline-K was obtained by the treatment of **11d** with potassium

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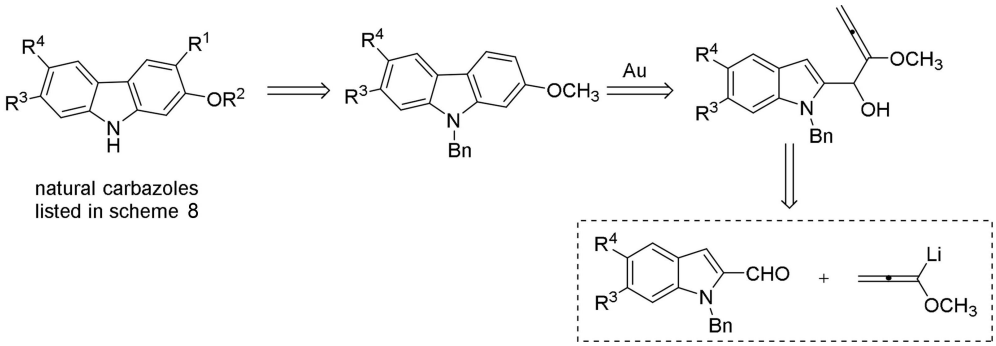


Scheme 8. Naturally occurring 2-oxygenated carbazole alkaloids.

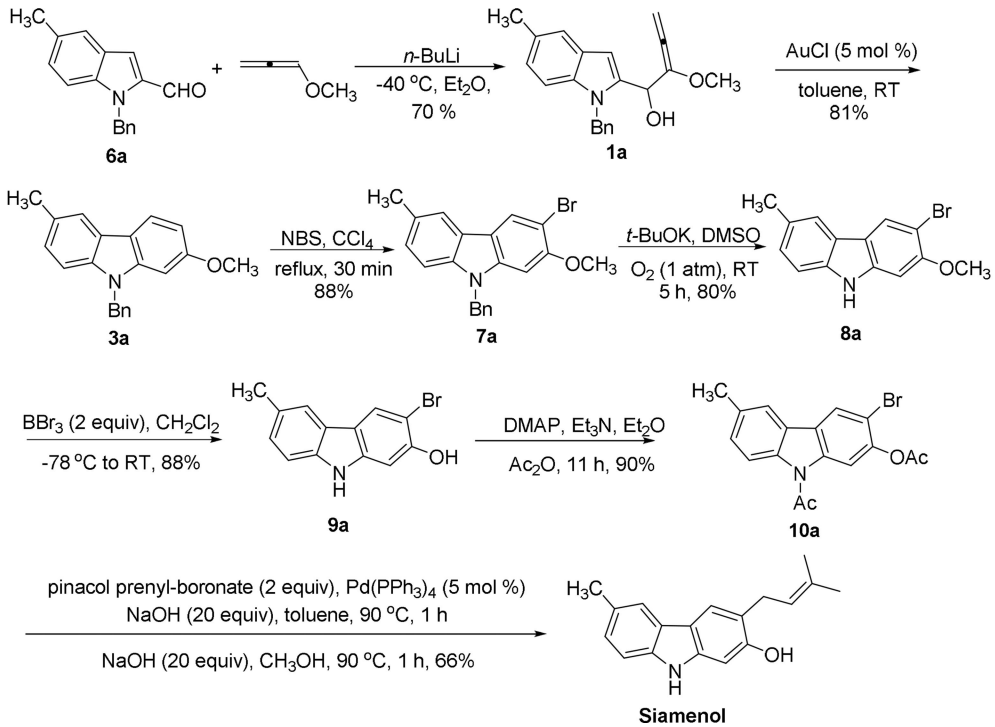
hydride, *t*-BuLi, and DMF to install the aldehyde functionality.[27] Clausine C was synthesized by methylation of the carboxylic acid in clausine-N with MeI. 3-Formyl-7-hydroxycarbazole and clausine-M were provided, respectively, by removal of the methyl group in clauszoline-K and clauszoline-L by treatment with BBr_3 in CH_2Cl_2 (Scheme 13).[26] Total synthesises of these carbazole alkaloids from methoxypropadiene are: clauszoline-K (4 steps with 22% overall yield), clauszoline-L (5 steps with 32% overall yield), 3-formyl-7-hydroxycarbazole (5 steps with 11% overall yield), clausine-M (6 steps with 19% overall yield), and clausine-N (4 steps with 34% overall yield) [15].

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Scheme 9. Retrosynthetic analysis of carbazole alkaloids listed in Scheme 8.



Scheme 10. Synthesis of siamenol.

3.1.5 Gram Scale Synthesis of 7-Methoxy-O-methylmukonal, Clausine-O, Clausine-K, Clausine-H, 7-Methoxymukonal, and Methyl 2-Hydroxy-7-methoxy-9H-carbazole-3-car-boxylate

In the continuous investigation of our group, the total synthesis of several natural carbazole alkaloids have been realized on gram scales via this efficient Au-catalyzed cyclization strategy, which shows the potential for its application as a general protocol for the practical synthesis of other naturally occurring carbazole alkaloids.[22]

A Large Scale Synthesis of 9-benzyl-2,7-dimethoxy-9H-carbazole ***3e***

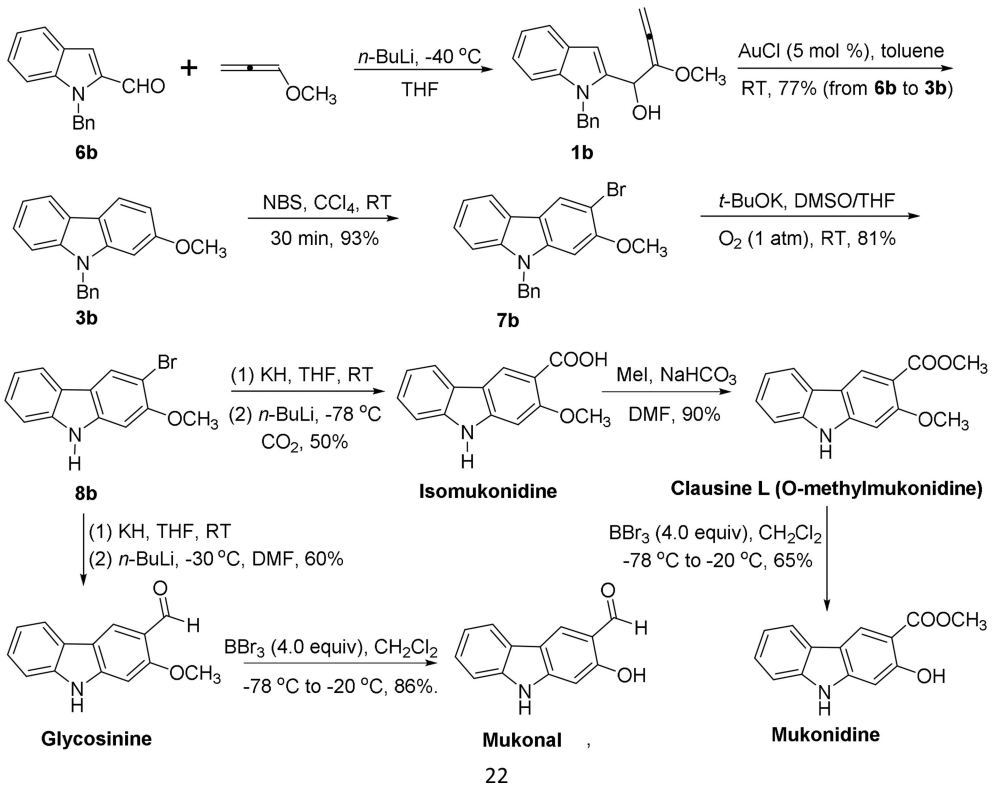
When the reaction scale for the synthesis of allenol **1e** was enlarged to 60 mmol, the reaction conditions need to be modified accordingly to make it more efficient. After several trials, the reaction on 60 mmol (15.9 g) was conducted with only 2% mol AuCl in the presence of 8.0 g of 4 Å MS to provide 17.4 g (91% isolated yield) of **3e** after simple recrystallization (Scheme 14).

Highly Selective Monobromination of 2,7-Dimethoxy-9H-carbazole

Due to the symmetric nature of 2,7-dimethoxy-9*H*-carbazole **3e**, selective monobromination at C-3 avoiding dibromination

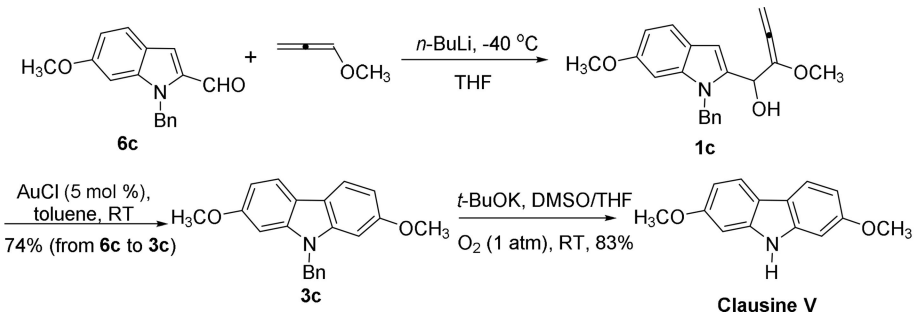
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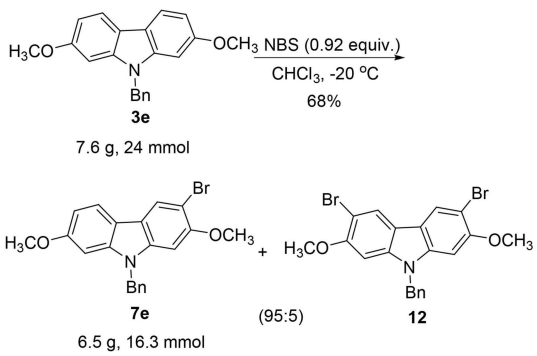
22

Scheme 11. Synthesis of isomukonidine, clausine L, mukonidine, glycosinine, and mukonal.



Scheme 12. Synthesis of clausine V.

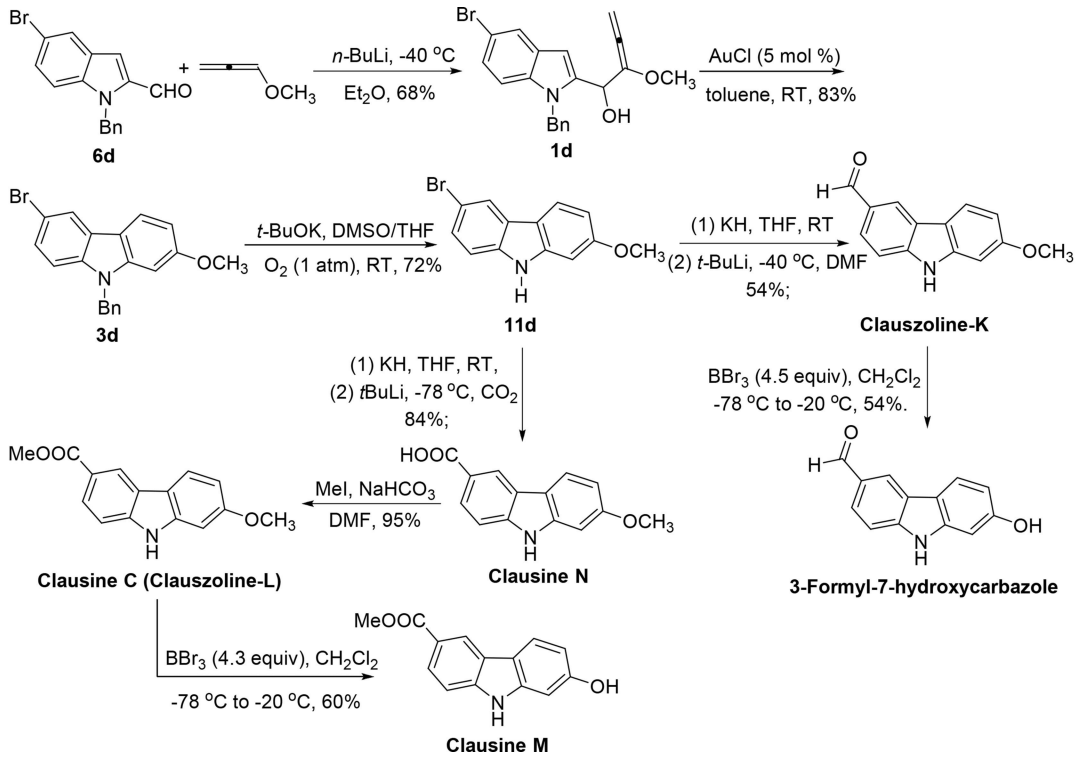
to form the byproduct **12** is challenging. After the optimization of reaction conditions, the best conditions of monobromination were defined as follows: bromination with 0.92 equiv. NBS in $CHCl_3$ at −20 °C, With the optimized conditions, we executed a reaction on 24 mmol (7.6 g) to afford **7e** (6.5 g, 16.3 mmol) in 68% yield with a ratio of **7e** and **12** as 95:5 as shown by NMR analysis (Equation 2).



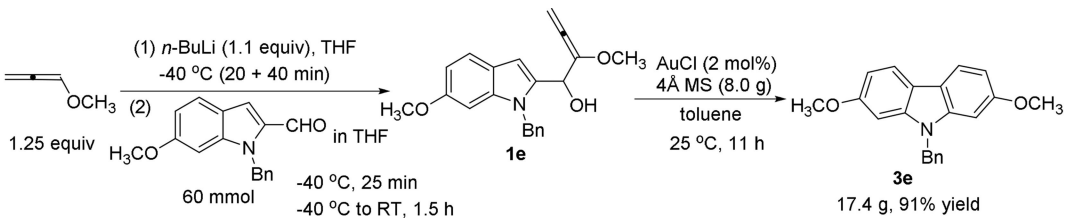
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Scheme 13. Synthesis of 3-formyl-7-hydroxycarbazole, clauszoline-K, clauszoline-L, clausine-M and clausine-N.



Scheme 14. A large scale synthesis of **3e**.

Gram Scale Total Synthesis

Deprotection of *N*-benzyl group of compound **7e** using *t*-BuOK/DMSO in O_2 atmosphere[25] afforded 3-bromo-2,7-dimethoxy-9*H*-carbazole **8e** in 77% yield. Compound **8e** is the key precursor for 7-methoxy-*O*-methylmukonal, clausine-K, clausine-H and clausine-O. 7-Methoxy-*O*-methylmukonal was obtained in 62% yield by treatment of **8e** with KH, *n*-BuLi, and DMF. Then the demethylation of both methyl ether units in 7-methoxy-*O*-methylmukonal with BBr_3 in CH_2Cl_2 provided clausine-O in 71% yield. Deprotonation of **8e** with KH, Br–Li exchange, and addition of dimethylformamide produced the target molecule clausine-K in 68% yield. Methylation of clausine-K with MeI generated clausine-H in 70% yield (Scheme 15).[22]

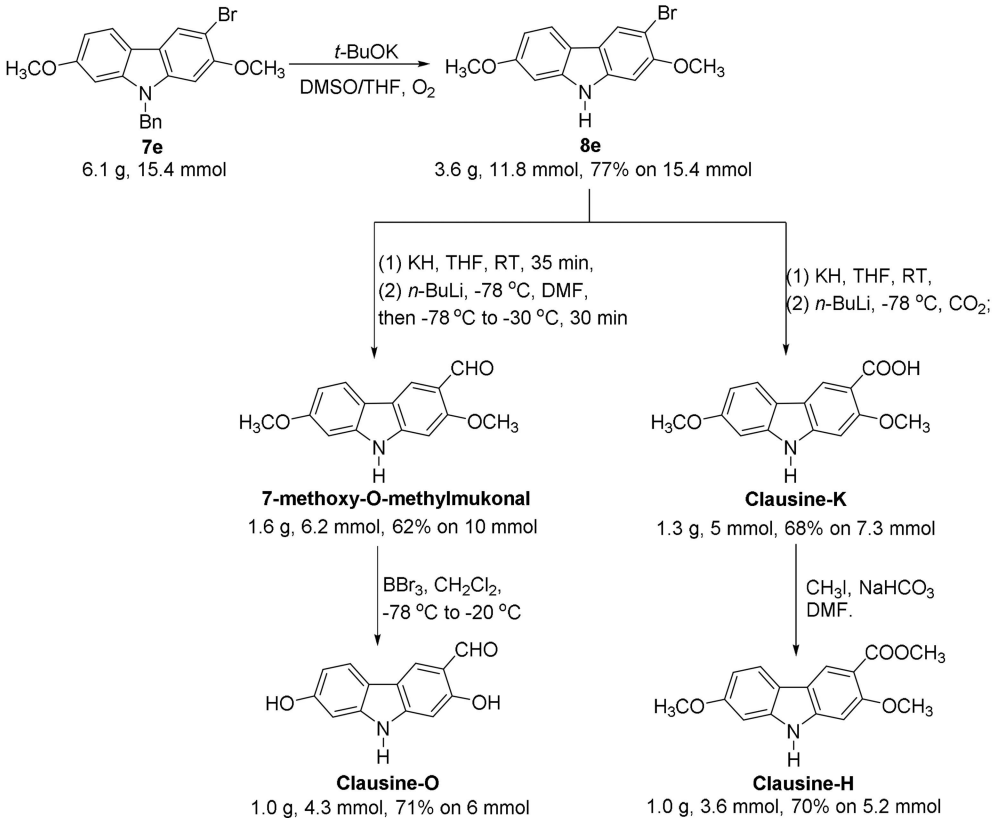
Selective removal of the methyl group at the C-2 position accomplishing the conversion of 7-methoxy-*O*-mukonal and clausine-H to 7-methoxymukonal and methyl 2-hydroxy-7-methoxy-9*H*-carbazole-3-carboxylate is another challenge. After systematic screening, we overcame the challenge by using $AlCl_3$, which could selectively cleave the etheric C–O bond adjacent to the carbonyl group by chelating with the oxygen atom of carbonyl group: 7-methoxymukonal and methyl 2-hydroxy-7-methoxy-9*H*-carbazole-3-carboxylate were synthesized in 94% yield and in 92% yield, respectively, as single products from the selective demethylation (Scheme 16).

3.1.6 Gram Scale Total Synthesis of 2-Oxygenated Carbazole Alkaloids: 2-Hydroxy-3-methylcarbazole, Pyrano[3,2-*a*] carbazole Alkaloids and Prenylcarbazole Alkaloids

According to the retrosynthetic analysis presented in Scheme 17,[23] the construction of the pyran ring and the installation of the prenyl group are the keys for the synthesis

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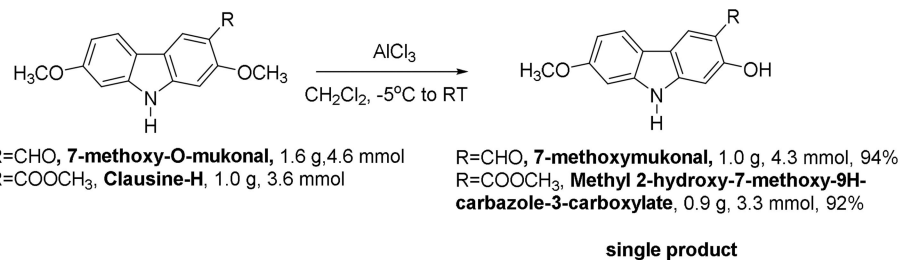


Scheme 15. Total synthesis of 7-methoxy-O-methylmukonal, causine-O, clausine-K (clauszoline-J), and clausine-H (clauszoline-C).

of these target pyrano[3,2-a]carbazole alkaloids (girinimbine and murrayacine) and prenylcarbazole alkaloids (mukoenine-A and heptaphylline), in addition to the carbazole skeleton formation. The strategies employed here to build the six-membered oxacycle in girinimbine and murrayacine involve Knoevenagel condensation of 9-benzyl-3-methyl-2-hydroxycarbazole **13b** with α,β-unsaturated aldehydes, followed by 6π-electrocyclization. The prenylation could be achieved via the Claisen rearrangement of corresponding *O*-2-methylbut-3-en-2-yl aryl ether **14**, which may be obtained by the palladium-catalyzed decarboxylative allylation of 3-methyl-2-hydroxycarbazole **13a**. Thus, 3-methyl-2-hydroxycarbazole **13** was taken as the key precursor for girinimbine, murrayacine, mukoenine-A, and heptaphylline syntheses.

*Synthesis of 9-Benzyl-2-methoxy-3-methylcarbazole **15***

Readily available methoxypropadiene and 1-benzylindole-2-carbaldehyde **6b** were taken as the starting materials to prepare the required indoly-2,3-allenol **1b**. Without purification, AuCl-catalyzed cyclization of allenol **1b** afforded 2-oxygented *N*-benzylated carbazole **3b** in the presence of 4 Å MS and 2 mol% AuCl in 89% yield. Subsequent regioselective electrophilic bromination of **3b** at C-3 with *N*-bromosuccinimide provided 9-benzyl-3-bromo-2-methoxycarbazole **7b** exclusively. 9-Benzyl-2-methoxy-3-methylcarbazole **15** was finally generated by treatment of **7b** with *n*-BuLi followed by methylation with methyl iodide. With such a sequence,



R=CHO, **7-methoxy-O-mukonal,** 1.6 g,4.6 mmol
R=COOCH3, **Clausine-H**, 1.0 g, 3.6 mmol

R=CHO, **7-methoxymukonal,** 1.0 g, 4.3 mmol, 94%
R=COOCH3, **Methyl 2-hydroxy-7-methoxy-9H-carbazole-3-carboxylate**, 0.9 g, 3.3 mmol, 92%

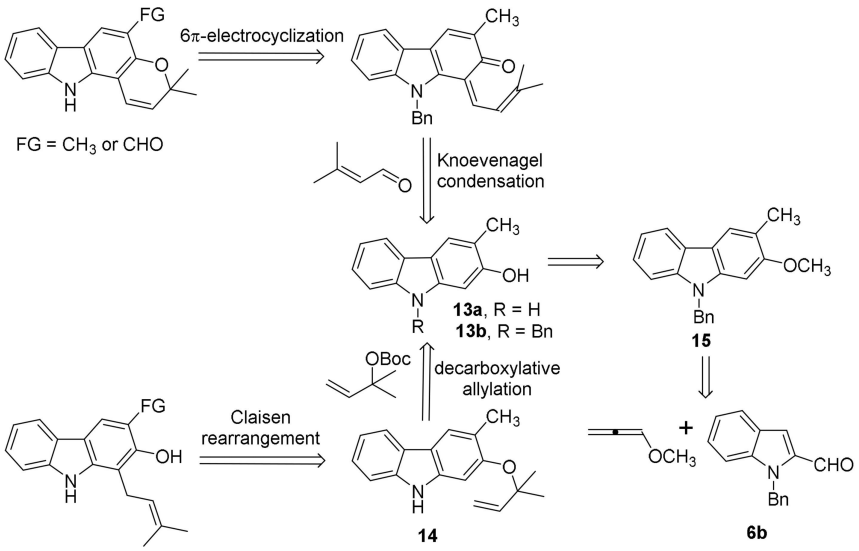
single product

Scheme 16. Total synthesis of 7-methoxymukonal and methyl 2-hydroxy-7-methoxy-9*H*-carbazole-3-carboxylate.

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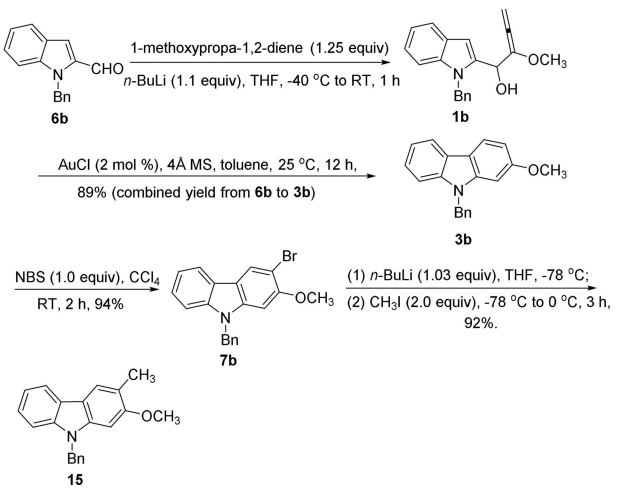


Scheme 17. Retrosynthetic analysis of girinimbine, murrayacine, mukoenine-A, and heptaphylline.

methylated carbazole **15** was synthesized in five steps with 78% overall yield from **6b** (Scheme 18).[23]

Synthesis of Girinimbine and Murrayacine

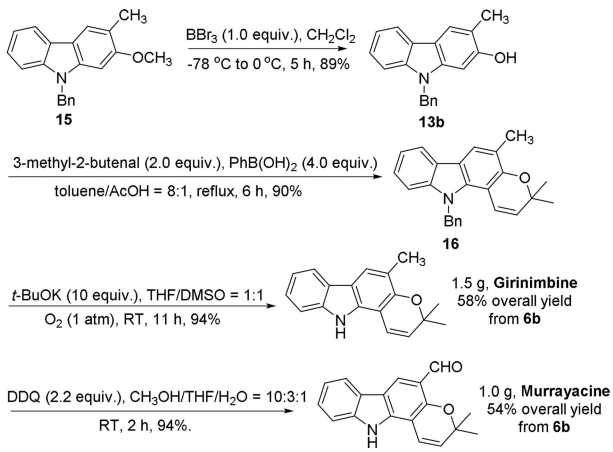
Demethylation of **15** with BBr_3 in CH_2Cl_2 provides 9-benzyl-3-methyl-2-hydroxycarbazole **13b** in 89% yield. The six-membered oxa-ring in **16** was successfully built in 90% isolated yield by the reaction of **13b** with 3-methylbut-2-enal using $PhB(OH)_2$ (4 equiv) in AcOH-toluene solution under reflux for 6 hours.[28] The final deprotection of *N*-benzyl group with excess potassium *tert*-butoxide/DMSO under oxygen atmosphere[25] could produce girinimbine with 58% overall yield form **6b**. Oxidation of girinimbine by using 2,3-dichloro-5,6-dicyano-1,4-benzoquinone (DDQ) as the oxidant provided murrayacine with 54% overall yield form **6b** (Scheme 19).



Scheme 18. Synthesis of carbazole **15**.

Synthesis of 2-Hydroxy-3-methylcarbazole, Mukoenine-A and Heptaphylline

From methylated carbazole **15**, the naturally occurring carbazole alkaloid 2-hydroxy-3-methylcarbazole **13a** could be afforded in an excellent yield by deprotection of *N*-benzyl group using excess *t*-BuOK/DMSO in O_2 atmosphere and subsequent demethylation with treatment of BBr_3 in CH_2Cl_2. The $Pd(PPh_3)_4$-catalyzed allylation of **13a** provided O-2-methylbut-3-en-2-ylated ether **18**. Then mukoenine-A could be afforded in high yield by the subsequent Claisen rearrangement of **18**. Considering the fact that the oxidant DDQ is

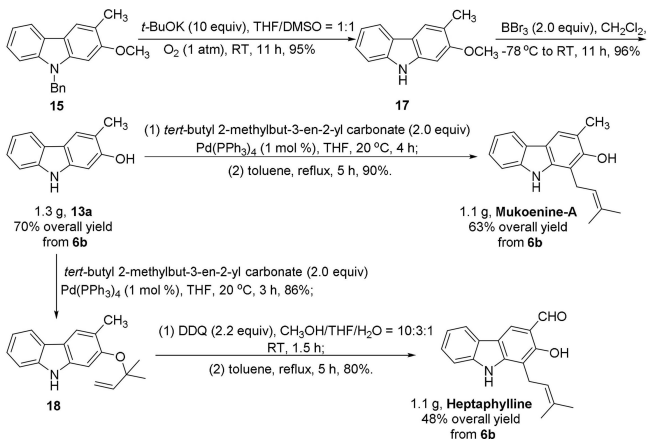


Scheme 19. Synthesis of girinimbine and murrayacine.

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Scheme 20. Synthesis of mukoenine-A and heptaphylline.

sensitive to hydroxyl group, heptaphylline was synthesized by the oxidation of **18** with DDQ and subsequent Claisen rearrangement in refluxing toluene (Scheme 20).[23]

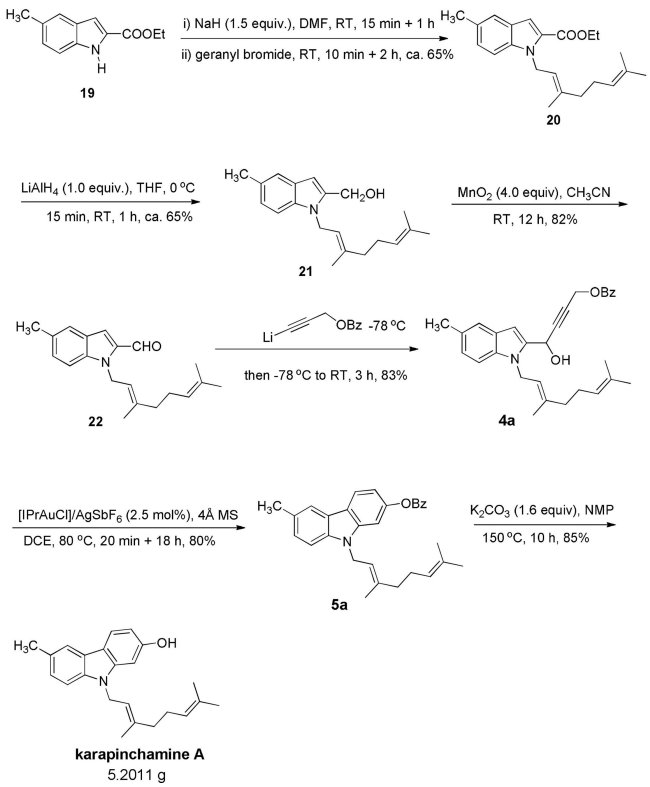
3.2 First Total Synthesis of Karapinchamine A

In 2013, karapinchamie A with a *N*-geranyl substituent[29] was isolated from *Murraya Koenigii*, which is a used as herb medicine in southern Asia, and showed useful melanogenesis inhibitory activity. [IPrAuCl]/$AgSbF_6$-catalyzed reaction of 4-benzoxyl-1-(indol-2-yl)-2-alkynols has a major advantage of its diversity, which allows even the *N*-substituent. Thus, the first total synthesis of karapinchamine A was executed by using the reaction presented in Table 3 as the key step.[20]

The commercially available ethyl 5-methyl-1H-indole-2-carboxylate **19** was treated with (2*E*)-geranyl bromide to introduce the geranyl group to the nitrogen atom. Subsequent reduction with $LiAlH_4$ and oxidation with MnO_2 afforded indole-carbaldehyde **22**. Then the reaction of **22** with (3-(benzoyloxy)prop-1-yn-1-yl)lithium provided the required propargylic alcohol **4a** in 83% yield conveniently. Under the catalysis of [IPrAuCl]/$AgSbF_6$ in the presence of 4 Å MS, carbazole **5a** could be produced exclusively in 80% yield. Eventually, karapinchamine A was afforded in 85% yield (5.2011 g) by the final deprotection of benzoates of **5a** with potassium carbonate (Scheme 21).[20]

4. Conclusion and Perspectives

In this account, we summarized our recent progress achieved in the studies of Au-catalyzed benzannulation and its application towards the synthesis of naturally occurring carbazole alkaloids. Due to the outstanding regioselectivity, atom economy, mild reaction conditions, scope, and functional group tolerance, these Au-catalyzed reactions would provide an efficient access to the related natural products and



Scheme 21. Synthesis of karapinchamie A.

unnatural products, which will be of interest to medicinal chemists. Further investigations including biological activity studies and other carbazole alkaloids synthesis are still being carried out in our laboratory.

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Received: April 22, 2017
Accepted: March 11, 2018
Published online on April 17, 2018

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